

Analytic State-Averaged DMRG-CASSCF Gradients and Nonadiabatic Couplings in a PySCF/pyblock2 Workflow

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Abstract

Analytic nuclear gradients and nonadiabatic coupling vectors are central ingredients for locating conical intersections and for propagating nonadiabatic dynamics with multireference electronic-structure methods. Complete-active-space self-consistent-field (CASSCF) theory provides a robust reference near electronic degeneracies, but full configuration interaction in the active space limits the accessible number of active orbitals. Here we report a PySCF/pyblock2 implementation of analytic state-averaged density matrix renormalization group CASSCF (SA-DMRG-CASSCF) gradients and nonadiabatic couplings, together with a SHARC-compatible interface for requesting energies, dipoles, gradients, and derivative couplings from the same workflow. The implementation is validated against PySCF full-CI active-space references on public benchmark systems. We further quantify the finite-bond-dimension response error by replacing FCI active-space vectors with block2 matrix product states at fixed FCI-optimized orbitals. The resulting convergence curves separate intrinsic DMRG compression error from artifacts associated with reoptimizing orbitals in a truncated CI manifold. An ethylene interface regression test demonstrates that the implemented DMRG-CASSCF backend writes all requested SHARC data blocks.

1 Introduction

Nonadiabatic molecular processes require electronic-structure information beyond adiabatic energies. Surface-hopping algorithms and related mixed quantum-classical methods use gradients and state couplings to propagate nuclear motion through regions where the Born-Oppenheimer separation breaks down.[1, 2, 3] For same-spin internal conversion, analytic derivative coupling vectors are especially useful because they define, together with state gradients, the first-order topology near conical intersections and provide direct inputs to optimization and dynamics workflows.[4]

CASSCF and state-averaged CASSCF (SA-CASSCF) wave functions are widely used for this purpose because they provide a balanced multiconfigurational description of near-degenerate states. The conventional active-space solver, however, is a full configuration interaction (FCI) expansion whose dimension grows combinatorially with active electrons and orbitals. The density matrix renormalization group (DMRG) replaces the FCI vector by a matrix product state (MPS), making compact approximations to much larger active spaces possible when the entanglement structure is favorable.[5, 6, 7]

Analytic response theory for DMRG-CASSCF has developed rapidly. Freitag and co-workers introduced an approximate SA-DMRG-SCF gradient and nonadiabatic coupling formalism based on single-site MPS Lagrange multipliers in a mixed-canonical representation.[8] Iino, Shiozaki, and Yanai later formulated analytic nuclear gradients for SA-DMRG-CASSCF using newly derived coupled-perturbed equations.[9] These developments establish the theoretical foundation for

DMRG-based derivative calculations. The remaining practical need addressed here is an open, reproducible PySCF-based implementation with regression tests, public benchmark data, and an interface usable by SHARC-style nonadiabatic dynamics workflows.

This work makes three contributions. First, we implement two complementary SA-DMRG-CASSCF response paths in a PySCF/pyblock2 environment: a native MPS response class and a PySCF-compatible active-space solver wrapper. Second, we validate gradients and nonadiabatic couplings against FCI active-space references and characterize the finite-bond-dimension response error on public benchmark systems. Third, we expose the validated backend through a SHARC-compatible PySCF interface and demonstrate the complete output path on a public ethylene test case.

2 Theory and Algorithm

2.1 State-Averaged CASSCF Response

For an SA-CASSCF wave function, the orbitals and active-space CI coefficients are optimized for a weighted average of several electronic states. Analytic gradients and nonadiabatic couplings are obtained by differentiating a Lagrangian that enforces stationarity with respect to orbital rotations and active-space wave-function parameters. The resulting coupled-perturbed CASSCF equations determine the orbital-response multipliers that enter the effective one- and two-particle density matrices used in the derivative contractions.[4, 8]

2.2 DMRG Active-Space Representation

In DMRG-CASSCF, each active-space state is represented by an MPS rather than by an explicit FCI vector. A finite bond dimension M controls the rank of the MPS approximation and therefore the accuracy of energies, density matrices, gradients, and derivative couplings. Following the SA-DMRG-SCF response strategy of Freitag et al., the present implementation evaluates the response quantities needed by PySCF’s SA-CASSCF derivative machinery while using pyblock2/block2 for the MPS representation and DMRG optimization. [8, 10]

2.3 Bond-Dimension Response Error

To isolate the effect of the MPS compression, we define a fixed-orbital bond-dimension response error. For each benchmark system we first converge a standard SA-CASSCF calculation with an FCI active-space solver. We then hold the converged molecular orbitals fixed, run spin-adapted block2 DMRG in the same active-space Hamiltonian at a selected bond dimension M , convert the resulting MPS roots into the PySCF active-space representation, and recompute analytic gradients and nonadiabatic couplings. The error is measured against the FCI active-space derivative at the same orbitals,

$$\Delta_g(M) = \left\| \nabla_{\mathbf{R}} E_0(M) - \nabla_{\mathbf{R}} E_0^{\text{FCI}} \right\|_2, \quad (1)$$

$$\Delta_d(M) = \min_{\sigma=\pm 1} \left\| \mathbf{d}_{01}(M) - \sigma \mathbf{d}_{01}^{\text{FCI}} \right\|_2, \quad (2)$$

where the sign minimization removes the arbitrary global phase of the electronic states. This protocol avoids mixing the MPS compression error with nonstationarity from orbital reoptimization on a deliberately truncated CI manifold.

3 Implementation

The implementation is built on PySCF[11, 12] and pyblock2/block2.[10] Two code paths are provided. The native `CPDMRG_CASSCF_ResponseMPS` path keeps the response operations in MPS form. The `MPSAsFCISolver` path exposes a PySCF-compatible active-space solver interface, allowing the same DMRG roots to be used by PySCF’s CASSCF gradient and nonadiabatic-coupling infrastructure. The latter path is the main regression-test target because it permits direct comparison with standard FCI active-space calculations.

The SHARC-facing interface adds a PySCF template method, `dmrg-casscf`. When requested, the interface constructs a PySCF CASSCF object, installs the DMRG-backed active-space solver, evaluates the requested Hamiltonian, dipole, gradient, and derivative-coupling quantities, and writes them in SHARC `QM.out` format. The ethylene example in this work is used as an end-to-end interface regression test.

4 Computational Details

All benchmark calculations used singlet state averaging over two roots with equal weights. The public systems are listed in Table 1. The bond-dimension response-error calculations used FCI-converged SA-CASSCF orbitals as the fixed orbital basis. Spin-adapted SU2 block2 DMRG was then run at each selected bond dimension, and the resulting state vectors were phase-aligned to the FCI reference before evaluating derivative errors. Gradients are reported in mE_h/Bohr and derivative couplings in atomic units.

Table 1: Public validation and benchmark systems.

System	Basis	Active space	Role
H ₄ chain	STO-3G	CAS(4,4)	Small-system response validation
H ₂ O	STO-3G	CAS(4,4)	Small-system response validation
N ₂	STO-3G	CAS(6,6)	Medium-active-space response validation
H ₂ O	6-31G	CAS(6,6)	NAC convergence benchmark
C ₂	STO-3G	CAS(8,8)	Larger active-space benchmark
LiF	STO-3G	CAS(4,4)	Nonzero-NAC benchmark
Ethylene	STO-3G	CAS(2,2)	SHARC-interface regression test

5 Results and Discussion

5.1 Validation Against FCI Active-Space References

The implementation is covered by unit and integration tests for DMRG/FCI energies, one- and transition-density matrices, MPS-to-FCI conversion, analytic gradients, analytic derivative couplings, and SHARC output formatting. The current regression suite contains 46 passing tests across 14 test files. These tests verify both direct numerical agreement with FCI active-space references in small active spaces and consistency between the native MPS and PySCF-compatible solver paths.

5.2 Finite-Bond-Dimension Response Error

Figure 1 shows the fixed-orbital bond-dimension response error. The H_4 , $H_2O/STO-3G$, N_2 , and $H_2O/6-31G$ curves show systematic decay of the derivative error as M is increased. The CAS(8,8) C_2 system is a more demanding benchmark; it improves substantially with M but also shows the expected sensitivity of excited-state tracking in a compact benchmark. LiF provides a nonzero derivative-coupling test case. Its energy and gradient errors converge rapidly, whereas the derivative coupling remains more gauge-sensitive, so it is interpreted as a diagnostic benchmark rather than the primary convergence example.

The key observation is that the catastrophic response spikes observed with an earlier SVD-truncated CI-vector proxy disappear when real DMRG roots are evaluated at FCI-stationary orbitals. This supports the interpretation that those spikes were not intrinsic failures of the DMRG response approximation, but artifacts of using a nonstationary truncated-CI orbital manifold.

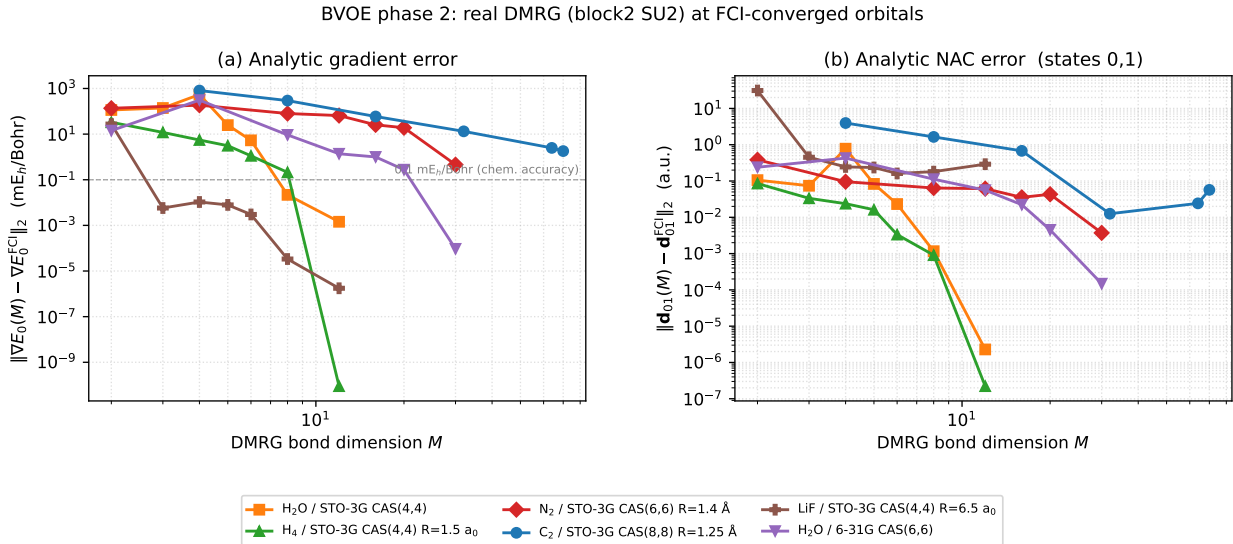


Figure 1: Fixed-orbital bond-dimension response-error convergence for public benchmark systems. Gradient errors are reported in mE_h/Bohr and nonadiabatic coupling errors in atomic units.

5.3 SHARC Interface Regression Test

The public ethylene test requests the Hamiltonian, dipole matrices, gradients for both singlet states, and the S_0/S_1 derivative coupling. The calculation completes with SHARC master error code 0 and writes all requested data blocks to `QM.out`. This confirms that the DMRG-backed PySCF object satisfies the data contract expected by the SHARC interface for energies, dipoles, gradients, and derivative couplings.

6 Conclusions

We have implemented analytic SA-DMRG-CASSCF gradients and nonadiabatic couplings in a PySCF/pyblock2 workflow and exposed the backend through a SHARC-compatible PySCF interface. The implementation is validated against FCI active-space references, and public benchmark data quantify how gradient and derivative-coupling errors decay with DMRG bond dimension at

fixed FCI-optimized orbitals. The results provide a reproducible open-source platform for testing DMRG-based multireference response quantities and for connecting them to nonadiabatic dynamics interfaces.

Data and Code Availability

The public code bundle contains the response implementation, tests, benchmark drivers, benchmark summaries, and plotting scripts. Application systems and trajectory data outside the public benchmark set are not included.

Acknowledgments

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